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The Title

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Abstract

Load balancing in RPL using Machine learning

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Chapter 1

Introduction

Recent development in the Internet of Things (IoT) have driven a rapid increase the number of devices connected to the Internet, with growth continues to accelerate in many areas: health care, construction, building.

Various projects need to deploy a large number of constrained devices, which are limited by computational power, memory and energy usage (battery), to cover large geographic areas. Because the underlying network technologies are limited in range, multi-hop routing solution is often required. Such networks are categorised as Low-power and Lossy Networks (LLN). Additionally, links in LLN are susceptible to loss and frequent packet drops.

These constrains led the Internet Engineering Task Force (IETF), which is an international organisation of network designers and researcher, to form Routing over Low-power and Lossy networks (ROLL) working group, investigating the routing solution for LLNs.

The ROLL first published RFC 5548 [4], RFC 5673 [11], RFC 5826 [3], RFC 5867 [9] to define the routing requirements for smart urban, industrial area, home automation and building automation applications respectively. Building on these requirements, ROLL proposed a new routing protocol designed specifically for LLN called IPv6 based routing protocol for low power and lossy networks (RPL), standardised in RFC 6550 [2]. Since its standardised, RPL has been employed in various settings and applications.

At its core, RPL constructs a Destination-Oriented Directed Acyclic Graph (DODAG) to route packets from nodes toward a root, using objective function (OF) to calculate path cost and select parents during DODAG construction. Because of this, OFs determine how the traffic is distributed across DODAG.

The protocol fell short when it comes to performing under load as load balancing was not considered in the design. Moreover, most of the OFs are reactive in nature, meaning that would react after a route has degraded, leading to decrease in network quality and reliability.

In this project, we propose a new machine learning based OF that take load balancing into account and predict the link quality to choose the parents

1.1 Contribution

- Contribute a new objective function based on machine learning which focuses on improving delivery ratio
- Investigate the feature importance of different metrics in determine the delivery ratio quality

1.2 Organisation

The rest of this report is structured as follow

- Chapter 2 provides background information in relation to RPL and objective functions and related work in improving the objective functions
- Chapter 3 provides the overview of the machine learning techniques and models used in this report
- Chapter 4 shows the experiment and result from applying ML to OF
- Chapter

Chapter 2

Background and Related Work

This chapter provides a overview of RPL and load balancing objective function in RPL in general.

2.1 Routing Protocol for Low-Power and Lossy Networks (RPL)

RPL is an oriented distance vector routing protocol, operating on top of IEEE 802.15.4 PHY and MAC-layer and over 6LoWPAN (IPv6 over Low power Wireless Personal Area Networks). Its main target is Wireless Sensor Networks (WSNs), where multipoint-to-point is the dominant communication pattern. Routes in RPL are optimised for traffic to or from one or more roots in the topology, while maintaining reliable and energy efficient data collection. To sustain this, RPL is designed adapt to various network condition, allowing nodes to switch route when route degradation is detected.

2.1.1 Destination-Oriented Directed Acyclic Graph (DODAG)

In principle, RPL organises nodes under DODAG, in which the sink node is usually the root node. When a node need to send packets to the sink node, it sends the packets to the chosen parent, which then forward to its parent. The process is repeated until the packets reach the sink node. DODAG is loop free, preventing packets getting stuck in endless circle.

In DODAG construction, each node identifies a set of potential nodes that have path to reach the DODAG root as candidates and determines one as its parent through objective function, which translate one or more metrics into rank and determine how to select the best parent. Rank in essence is the estimation of the distance between current node and the root node where only nodes with higher rank are considered as potential parent.

RPL offers flexibility by separating packet processing and forwarding from routing optimisation, leaving the OF to be specified depends on application, e.g. lowering energy usage, lowering latency,...

There are four identifiers to define maintain a topology in RPL Table 2.1, with RPLInstanceID and DODAGID used to identify a single DODAG in the network.

To form and control DODAG, RPL adds five types of control messages as specified in ICMPv6 via RFC 6550 [2]

- DODAG Information Object (DIO): carries information that allow nodes to discover a RPL instance, select parents and maintain the DODAG.
- DODAG Information Solicitation (DIS):
- Destination Advertisement Object (DAO)

Table 2.1: RPL DODAG identifiers

Identifier	Description
RPLInstanceID	The identity of a RPL instance, which may contain multiple DODAGs optimised for different OFs and applications.
DODAGID	Specifies a DODAG within a RPL instance.
DODAGVersionNumber	When a DODAG rebuild is needed, the root node increments the version number.
Rank	Defines the position of a node relative to the DODAG root.

- Destination Advertisement Object Acknowledgement (DAO-ACK)
- CC

The root starts the DODAG formation by sending DIO to its neighbouring nodes. Upon receiving DIO, a node decide whether to join the advertised DODAG or not. To prevent loops, a node only considers a DIO if it advertises a rank lower than its own rank. In certain implementation, a node would wait for several DIOs to build the pool of candidate parents before deciding. For each candidate, the node computes the path cost of the corresponding link and selects the candidate with the lowest path cost as its preferred parent.

Once a parent has been selected, the node broadcasts its own DIO to advertise itself to its neighbours. This process is done until all the nodes have join the DODAG.

A node would send DAO message back to the root node after it has chosen a parent, Afterward, the nodes broadcast its DIO to advertise itself to the neighbour nodes. A node only consider the DIO if it has ranks lower than the rank advertised in the DIO. In the DAO-ACK mode, a node wait for the root node to send back DAO-ACK before confirming that it is reachable and advertise DIO to surrounding nodes

RPL also uses DIS when a new node want to join the network. Upon receiving DIS, nodes will unicast their DIO to the sending node

2.1.2 Trickle Timer

In addition to that, RPL also employs trickle timer algorithm to determine when to send DIO. Each DIO send will double the interval up until INTERVAL_MAX. When a node switch parents, it also resets the DIO interval and timer

2.1.3 Communication Patterns

RPL supports upward and downward route and point-to-point communication. The upward route is constructed from DODAG. For downward route, RPL supports two modes non-storing and storing mode. In non-storing, the packet is sent to the sink node. The sink node would then calculate the downward path and route them to the destined node. In storing mode, each node would store its own routing table and forward to the node in the table.

The proposed OFs by IETF presented in have defined some recommendation on how to use a metrics. There are two default implementation of OF call OF0 and MRHOF.

2.2 Review of RPL Objective Functions

2.2.1 Objective Function zero (OF0)

2.2.2 Minimum Rank with Hysteresis Objective Function (MRHOF)

Final piece of RPL is objective function, which is used to determine the path cost of a given link. There are two default implementations of OF: OF0 (Objective Function 0) and MRHOF. In OF0, hop count is used to determine the path cost, a node would switch parent as soon as it receives a better parent DIO. This would lead to more parent switch and potentially less stable DODAG. In contrast, MRHOF proposes multiple thresholds before switching, in particular path cost threshold and time threshold. In path cost threshold, unless the potential node is better than the current parent by a threshold, it will stick with the current parent. However, after the time threshold, if that node is still better then the node will switch parent. This would allow a more stabilised DODAG. Additionally, MRHOF uses link metric ETX instead of hop count. Several metrics have been proposed to be used in objective functions. However, the specification of RPL leaves the choice of which metrics to use open to the implementation.

2.2.3 Related Work

Various objective functions have been proposed, which can be categorised into 4 groups: single metric, composite and context aware.

In [8], the authors used queue utilisation metric in objective function, called QU-RPL to deal with dropped packets due to queue overflow. However, only queue metrics are taken into consideration, without mentioning latency.

A new objective function: OF-FL is presented in [7], which employs fuzzy logic with four metrics.

- Various objective functions have been proposed - Types of Objective function - Single metrics - Composite - Context aware

- Energy aware: [1] - CLRPL: [13]

In recent, there has been many attempts to employ machine learning into OFs. Despite that, most of the new application of ML in RPL focuses on reinforcement learning with little focus on supervised learning

In [6], the author puts Q-Learning to OF, award the model when a packet is transmitted successfully and penalise when choosing a link that lead to packet loss. [5] utilised Random Weights with six metrics (...) and [10], [12] proposed using CatBoost models with eight metrics to predict the probability if a packet is transmitted successfully.

2.3 Contiki-NG and COOJA Simulator

2.3.1 Contiki

Contiki-NG is a lightweight operating system for resource-constrained devices in WSNs and IoT.

which was built on the Contiki.

The Swedish Institute of Computer Science (SICS) released the first version of Contiki in 2003. Contiki was created by Adam Dunkels in 2002, which supports various protocols for IoT devices and have been supported from various research institutions

However, recently a new Contiki cross-platform has been released on Nov 2017 named Contiki-NG 4.0, which is licensed under the 3-clause BSD license. Contiki-NG concentrates on

the Next-Generation IoT devices such as ARM Cortex M3 and other 32-bit MCUs and on reliable and secure standard protocols like IPv6/6LoWPAN, 6TiSCH, RPL, and CoA

In the next generation, the RPL implementation was rewritten to reduce the number of features and prune the unneeded feature, choosing non-storing mode as default and remove multi-DODAG functionality.

Additionally, ContikiMAC was removed in favor of TSCH implementation, thus making TSCH the default for duty cycle MAC implementation.

2.3.2 COOJA Simulator for Contiki-NG

The Cooja simulator is widely used among RPL OF researches

Cooja is an elastic Java-based simulator dedicated for WSNs based on Contiki Operating System. Cooja is capable of simulating different node types, including simulated hardware (consisting of clock cycle and MCU) and software based simulation (which skip the internal working of an MCU).

In this regard, Cooja is a tool for real hardware simulation.

depends on the mote type, provides three mote types: Sky, Z1 and Cooja.

Thus to maximise the computing power, Z1 is the most suitable mote for our need.

Chapter 3

Machine Learning

In this chapter, I will discuss about the machine learning models in IoT devices

3.1 Regression Machine Learning Model

The advance of machine learning has led the wide applications in IoT devices.

KNNR: works by comparing a node with the surrounding nodes to predict a value. In a sense, KNNR is considered a lazy model, in which it remembers all training data, thus result in a heavy model

Gaussian Process Regressor, similar to SVM. However, in [?], the authors analyse GPR and conclude that the memory requirement of GPR is $O(n^2)$ while the computation cost is $O(n^3)$, which is quite computational and memory expensive for constrained devices

Deep learning is recently a new advances and has achieve in terms of matching the complex patterns and data. However, due to the constrained environment of the working MCU which has several KB of RAM and ROM. I rule out this.

Random forest and Decision Tree operates on similar algorithm. In tree base algorithm, each value is split into left and right, following the path to reach the end. In Random Forest, multiple trees are used to prevent overfitting and provide better performance. However, in my experiments, the performance of random forest and decision tree are quite similar to each other. Thus I remove random forest from list of considered models as they are much heavier than decision tree.

Chapter 4

Experiments

4.1 Methodology

The OFs are simulated in COOJA, using Z1 mote Table ... list the running parameters for the experiment

Chapter 5

Evaluation

PDR Latency CPU util Parent switch
Model overhead (network and memory)

Chapter 6

Conclusions

The conclusions are presented in this Chapter.

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